

Section 19: IGP Fundamentals

Layer: L3 | Reference: Odom Vol1 Ch21: OSPF Concepts (IGP framing) | *Original study notes* — CCNA 200-301

Interior Gateway Protocols (IGPs) handle routing within a single autonomous system. This section covers the conceptual framing the CCNA exam expects before diving into OSPF specifics: classful vs classless behavior, protocol selection logic, and basic verification.

Key Concepts

- IGP = routing within one autonomous system (OSPF, EIGRP, RIP). Distinguished from EGP (BGP), which routes between autonomous systems
- Classful routing protocols (RIPv1) do not include subnet mask information in their updates and cannot support VLSM or discontinuous networks
- Classless routing protocols (RIPv2, OSPF, EIGRP) include subnet mask info in updates, fully supporting VLSM and CIDR
- Within an organization, the choice of IGP affects convergence speed, scalability, and multi-vendor compatibility (OSPF is an open standard; EIGRP was historically Cisco-proprietary)
- Administrative Distance, not metric, is what lets a router choose between two different IGPs advertising the same destination
- Metric is what lets a router choose between two paths within the SAME IGP
- Passive-interface is the standard way to keep a routing protocol from forming neighbor relationships on end-user-facing interfaces while still advertising that subnet
- 'show ip protocols' is the fastest way to verify which IGP is running, its timers, and which networks it's advertising

Command Reference

Command	Purpose
<code>show ip protocols</code>	Verify active routing protocols, timers, networks advertised, and passive interfaces
<code>router ospf <id> / router eigrp <as></code>	Enter the configuration mode for the chosen IGP
<code>network <ip> <wildcard mask> [area <id>]</code>	Advertise a network into the IGP (syntax detail varies by protocol)
<code>passive-interface default</code>	Make all interfaces passive by default; then selectively un-suppress real router-facing links
<code>no passive-interface <intf></code>	Re-enable active hellos on a specific interface after a 'passive-interface default'

Configuration Walkthrough

Pattern: passive by default, active only where routers actually peer

```
R1(config)#router ospf 1
R1(config-router)#passive-interface default
R1(config-router)#no passive-interface GigabitEthernet0/0
! Gi0/0 faces another router; all other OSPF-enabled interfaces stay passive
```

Worked Scenario

Q: A router is advertising a LAN subnet correctly via OSPF, but you notice unnecessary OSPF Hello traffic flooding that LAN segment toward end-user PCs. How do you clean this up?

Mark the LAN-facing interface as passive. The subnet still gets advertised into OSPF (so remote routers learn the route), but the router stops sending Hello packets out that interface, since there are no other routers there to form an adjacency with. This reduces unnecessary traffic and slightly improves security by not exposing routing protocol chatter to end-user segments.

Common Mistakes / Gotchas

- Believing passive-interface stops a network from being advertised - it doesn't; it only suppresses Hellos
- Forgetting that AD determines IGP-vs-IGP preference while metric determines path preference inside one IGP

Exam Tips

- ★ *If a question mentions 'between autonomous systems,' the answer involves EGP/BGP, not an IGP*
- ★ *Passive-interface is a recurring config pattern across OSPF and EIGRP - know the syntax variations*